

Data Scientist with 3+ years of experience engineering production-grade models from raw operational data to drive business value. Proven expertise in live industrial time-series forecasting, anomaly detection, and deploying LLM-based architectures like RAG and SLMs. Successfully owns the full MLOps lifecycle on AWS and Azure, partnering with domain experts to translate complex signals into live, monitored systems that improve operational efficiency.

PROFESSIONAL EXPERIENCE

Data Scientist (Thesis Researcher)

Metso

11/2025 — 06/2026

Frankfurt am Main, Germany

- Extended a **Graph Deviation Network** with **multi-head attention** and a **prior-informed graph structure**, benchmarked against streaming **PCA** and **Adaptive-PCA** for anomaly detection on **multivariate industrial time-series**, detecting **17 true anomalous events** missed by existing threshold monitoring.
- Built **Dockerised** training and inference pipelines that turned raw live sensor feeds from **50+ tags** into model-ready features with **CI/CD on Azure**, and shipped a **Grafana** root-cause dashboard that process engineers used during **live equipment testing** to trace alerts to responsible sensor groups.

Data Scientist (Working Student)

Metso

04/2025 — 10/2025

Frankfurt am Main, Germany

- Built a **real-time monitoring system** processing **100+ live process signals**, integrating outlier detection (**Isolation Forest**, **LOF**) and forecasting (**ARIMA**, **Prophet**, **LSTM**) to catch drift and early faults ahead of the plant's threshold alarms.
- collaborated with **domain experts** to cross-check predictions against real plant behaviour, translating their process knowledge into feature engineering and detection rules that cut false alarms by **40%**.

Data Science Intern

Metso

10/2024 — 03/2025

Frankfurt am Main, Germany

- Integrated the **ACT process control software** over **gRPC** to pull live process data, and built the plant's first **Grafana KPI dashboard**, pairing domain-informed health metrics with **linear regression** on sensor trends to give engineers a single view of equipment condition.

Data Analyst

Dreamwarez

08/2021 — 02/2023

Pune, India

- Built and maintained executive **Power BI** dashboards with **DAX-based KPIs**, consolidating fragmented multi-source operational data into a single reporting layer that replaced manual spreadsheet reporting for senior stakeholders.
- Shipped a **hospitality revenue analytics platform** for an e-commerce hotel client, delivering **RevPAR**, **GOPPAR**, **ADR**, and direct booking ratio on **Python and SQL pipelines** built from scratch, giving the client the visibility to move from fixed rates to **dynamic pricing**.

Software Developer Intern

Dreamwarez

02/2021 — 07/2021

Pune, India

- Developed responsive web applications with input validation and backend stability improvements, working within Agile/Scrum delivery cycles.

PERSONAL PROJECTS

Medical SLM | [Link](#)

07/2026

- Engineered a **60M-parameter** transformer from scratch in **PyTorch** by hand-writing advanced architectures (**multi-head latent attention**, **RoPE**, **RMSNorm**, **SwiGLU**), and successfully optimized pre-training on **600M+ tokens** under strict consumer GPU constraints.
- Orchestrated the end-to-end LLM lifecycle independently, from **parquet streaming data ingestion** and tokenizer training to **instruction tuning** and creating a custom zero-shot evaluation harness for MedMCQA and PubMedQA.

Enterprise Agentic RAG | [Link](#)

06/2026

- Developed a production-grade, multi-agent **RAG** system utilizing **LangGraph** and **FastAPI**, with **Qdrant** vector storage and Groq inference, successfully implementing multi-step reasoning and stateful **conversation memory** for enterprise document datasets.
- Engineered a secure, high-precision retrieval pipeline featuring semantic chunking, **FlashRank reranking**, and **history-aware query planning**, integrating **NeMo Guardrails** for jailbreak prevention and **Logfire**, **LangSmith**, and **RAGAS** for end-to-end tracing and evaluation.

Churn Prediction MLOps | [Link](#)

02/2026

- Engineered and productionized an end-to-end **XGBoost** churn model on **AWS**, integrating **Great Expectations** for input validation, **MLflow** for tracking, and **GitHub Actions** for **CI/CD** automation. Containerized the deployment using **Docker** and **FastAPI**, while establishing real-time reliability monitoring via **CloudWatch**.

EDUCATION

M.Sc. in Artificial Intelligence

Brandenburgische Technische Universität

Thesis : Unsupervised anomaly detection on non-stationary industrial multivariate time-series.

04/2023 — 08/2026

Cottbus, Germany

B.Tech. in Computer Science and Engineering

Lovely Professional University

06/2017 — 06/2021

Jalandhar, India

SKILLS

Languages & Frameworks	Python, SQL, PyTorch, FastAPI, gRPC, Git
Machine Learning & Data	Scikit-learn, XGBoost, Time-Series Forecasting, Anomaly Detection, Pandas, NumPy, Statsmodel
AI Engineering	LLMs/SLMs, Finetuning, LangGraph, Agentic RAG, Vector Databases, AI Guardrails, Hugging Face
MLOps & Infrastructure	Docker, MLflow, AWS, Azure, GitHub Actions, CI/CD
Evaluation & Observability	LangSmith, Logfire, RAGAS, Grafana, Plotly

PUBLICATION

Passenger Security Using Facial Emotion Recognition System | [Link](#)

11/ 2022

International Journal of Advance Research in Engineering, Science & Management (IJARESM)

LANGUAGES

English	Professional proficiency (C1)
Deutsch	Elementary proficiency (A2, actively improving)

ACHIEVEMENTS & INTERESTS

Secured **All India Rank 2,757** out of 157,000+ candidates in **GATE (Graduate Aptitude Test in Engineering)**, placing in the **Top 1.76%** (98.24th percentile) nationwide.

Secured **dual Gold Medals** in American Football and Rugby at the National Championships.

Perform live music (vocals, guitar, drums) at regional events and social gatherings on weekends.



Atharva Hatekar

Cottbus, 5 August 2026